$$C_{Hu}{}^{i1\_i2\_} \to$$
$$\frac{1}{16\pi^2}\left(-\frac{1}{216}\frac{1}{m_\Phi^2}\, y_u^{pi2}\left(7\, g_1^2\, c_\gamma^2\, \overline{y_u}^{pi1} + 18\, s_\gamma^2\left(\overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{si1}\left(6\, c_\gamma^2 + s_\gamma^2\right) + c_\gamma^2\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{rs}\right)\right) + \right.$$
$$\frac{4}{81}\sum_p g_1^4\, LF_{3,0}\left[m_{\tilde d}^{p}\right]\delta_{i1i2} - \frac{5}{54}\sum_p g_1^4\, LF_{4,-1}\left[m_{\tilde d}^{p}\right]\delta_{i1i2} + \frac{16}{405}\sum_p g_1^4\, LF_{5,-2}\left[m_{\tilde d}^{p}\right]\delta_{i1i2} +$$
$$\frac{4}{27}\sum_p g_1^4\, LF_{3,0}\left[m_{\tilde e}^{p}\right]\delta_{i1i2} - \frac{5}{18}\sum_p g_1^4\, LF_{4,-1}\left[m_{\tilde e}^{p}\right]\delta_{i1i2} + \frac{16}{135}\sum_p g_1^4\, LF_{5,-2}\left[m_{\tilde e}^{p}\right]\delta_{i1i2} +$$
$$\frac{2}{27}\sum_p g_1^4\, LF_{3,0}\left[m_{\tilde l}^{p}\right]\delta_{i1i2} - \frac{5}{36}\sum_p g_1^4\, LF_{4,-1}\left[m_{\tilde l}^{p}\right]\delta_{i1i2} + \frac{8}{135}\sum_p g_1^4\, LF_{5,-2}\left[m_{\tilde l}^{p}\right]\delta_{i1i2} +$$
$$\frac{2}{81}\sum_p g_1^4\, LF_{3,0}\left[m_{\tilde q}^{p}\right]\delta_{i1i2} - \frac{5}{108}\sum_p g_1^4\, LF_{4,-1}\left[m_{\tilde q}^{p}\right]\delta_{i1i2} + \frac{8}{405}\sum_p g_1^4\, LF_{5,-2}\left[m_{\tilde q}^{p}\right]\delta_{i1i2} +$$
$$\frac{16}{81}\sum_p g_1^4\, LF_{3,0}\left[m_{\tilde u}^{p}\right]\delta_{i1i2} - \frac{10}{27}\sum_p g_1^4\, LF_{4,-1}\left[m_{\tilde u}^{p}\right]\delta_{i1i2} + \frac{64}{405}\sum_p g_1^4\, LF_{5,-2}\left[m_{\tilde u}^{p}\right]\delta_{i1i2} +$$
$$\frac{1}{18}\, c_\gamma^2\, y_u^{pi2}\left(-g_1^2\, \overline{y_u}^{pi1} + 9\, c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{si1} - 9\, s_\gamma^2\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{rs}\right) LF_{1,2}[m_\Phi] -$$
$$\frac{1}{12}\, g_1^2\, c_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1}[m_\Phi] + \frac{1}{108}\, g_1^2\left(9\, c_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2} + 8\, g_1^2\, \delta_{i1i2}\right) LF_{3,0}[m_\Phi] -$$
$$\frac{5}{36}\, g_1^4\, LF_{4,-1}[m_\Phi]\,\delta_{i1i2} + \frac{8}{135}\, g_1^4\, LF_{5,-2}[m_\Phi]\,\delta_{i1i2} + \frac{2}{27}\, g_1^4\, LF_{3,0}[\tilde\mu]\,\delta_{i1i2} +$$
$$\frac{1}{9}\, g_1^4\, LF_{4,-1}[\tilde\mu]\,\delta_{i1i2} - \frac{16}{135}\, g_1^4\, LF_{5,-2}[\tilde\mu]\,\delta_{i1i2} + \frac{1}{36}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,0}\left[m_1, m_{\tilde q}^{p}\right] -$$
$$\frac{1}{18}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,-1}\left[m_1, m_{\tilde q}^{p}\right] + \frac{1}{36}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{4,1,-2}\left[m_1, m_{\tilde q}^{p}\right] +$$
$$\frac{2}{81}\, g_1^4\, LF_{2,1,0}\left[m_1, m_{\tilde u}^{i1}\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{2,2,-1}\left[m_1, m_{\tilde u}^{i1}\right]\delta_{i1i2} - \frac{4}{81}\, g_1^4\, LF_{3,1,-1}\left[m_1, m_{\tilde u}^{i1}\right]\delta_{i1i2} +$$
$$\frac{2}{81}\, g_1^4\, LF_{4,1,-2}\left[m_1, m_{\tilde u}^{i1}\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{2,1,0}\left[m_1, m_{\tilde u}^{i2}\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{2,2,-1}\left[m_1, m_{\tilde u}^{i2}\right]\delta_{i1i2} -$$
$$\frac{4}{81}\, g_1^4\, LF_{3,1,-1}\left[m_1, m_{\tilde u}^{i2}\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{4,1,-2}\left[m_1, m_{\tilde u}^{i2}\right]\delta_{i1i2} +$$
$$\frac{3}{4}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,0}\left[m_2, m_{\tilde q}^{p}\right] - \frac{3}{2}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,-1}\left[m_2, m_{\tilde q}^{p}\right] +$$
$$\frac{3}{4}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{4,1,-2}\left[m_2, m_{\tilde q}^{p}\right] + \frac{4}{3}\, g_3^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,0}\left[m_3, m_{\tilde q}^{p}\right] -$$
$$\frac{8}{3}\, g_3^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,-1}\left[m_3, m_{\tilde q}^{p}\right] + \frac{4}{3}\, g_3^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{4,1,-2}\left[m_3, m_{\tilde q}^{p}\right] +$$
$$\frac{2}{27}\, g_1^2\, g_3^2\, LF_{2,1,0}\left[m_3, m_{\tilde u}^{i1}\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{2,2,-1}\left[m_3, m_{\tilde u}^{i1}\right]\delta_{i1i2} -$$
$$\frac{4}{27}\, g_1^2\, g_3^2\, LF_{3,1,-1}\left[m_3, m_{\tilde u}^{i1}\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{4,1,-2}\left[m_3, m_{\tilde u}^{i1}\right]\delta_{i1i2} +$$
$$\frac{2}{27}\, g_1^2\, g_3^2\, LF_{2,1,0}\left[m_3, m_{\tilde u}^{i2}\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{2,2,-1}\left[m_3, m_{\tilde u}^{i2}\right]\delta_{i1i2} -$$
$$\frac{4}{27}\, g_1^2\, g_3^2\, LF_{3,1,-1}\left[m_3, m_{\tilde u}^{i2}\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{4,1,-2}\left[m_3, m_{\tilde u}^{i2}\right]\delta_{i1i2} +$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{2,2,0}\left[m_{\tilde d}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} -$$
$$\frac{1}{9}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{3,2,-1}\left[m_{\tilde d}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} +$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{2,2,0}\left[m_{\tilde e}^{r}, m_{\tilde l}^{p}\right]\delta_{i1i2} -$$
$$\frac{1}{9}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{3,2,-1}\left[m_{\tilde e}^{r}, m_{\tilde l}^{p}\right]\delta_{i1i2} -$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{3,1,0}\left[m_{\tilde l}^{p}, m_{\tilde e}^{r}\right]\delta_{i1i2} -$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{3,2,-1}\left[m_{\tilde l}^{p}, m_{\tilde e}^{r}\right]\delta_{i1i2} +$$
$$\frac{1}{2}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{4,1,-1}\left[m_{\tilde l}^{p}, m_{\tilde e}^{r}\right]\delta_{i1i2} -$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_e}^{pr}\right)\left(c_\gamma\, a_e^{pr} - s_\gamma\, \tilde\mu\, y_e^{pr}\right) LF_{5,1,-2}\left[m_{\tilde l}^{p}, m_{\tilde e}^{r}\right]\delta_{i1i2} -$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{3,1,0}\left[m_{\tilde q}^{p}, m_{\tilde d}^{r}\right]\delta_{i1i2} -$$
$$\frac{2}{9}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{3,2,-1}\left[m_{\tilde q}^{p}, m_{\tilde d}^{r}\right]\delta_{i1i2} +$$
$$\frac{5}{6}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{4,1,-1}\left[m_{\tilde q}^{p}, m_{\tilde d}^{r}\right]\delta_{i1i2} -$$
$$\frac{2}{3}\, g_1^2\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{pr} - s_\gamma\, \tilde\mu\, y_d^{pr}\right) LF_{5,1,-2}\left[m_{\tilde q}^{p}, m_{\tilde d}^{r}\right]\delta_{i1i2} -$$
$$\frac{1}{9}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{2,2,0}\left[m_{\tilde q}^{p}, m_{\tilde u}^{r}\right]\delta_{i1i2} +$$
$$\frac{1}{18}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{3,2,-1}\left[m_{\tilde q}^{p}, m_{\tilde u}^{r}\right]\delta_{i1i2} -$$
$$\frac{1}{18}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,0}\left[m_{\tilde q}^{p}, \tilde\mu\right] + \frac{1}{36}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,2,-1}\left[m_{\tilde q}^{p}, \tilde\mu\right] + \frac{1}{36}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}$$
$$LF_{3,1,-1}\left[m_{\tilde q}^{p}, \tilde\mu\right] + \frac{1}{9}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{3,1,0}\left[m_{\tilde u}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} +$$
$$\frac{1}{9}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{3,2,-1}\left[m_{\tilde u}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} +$$
$$\frac{1}{3}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{4,1,-1}\left[m_{\tilde u}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} -$$
$$\frac{2}{3}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\right)\left(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\right) LF_{5,1,-2}\left[m_{\tilde u}^{r}, m_{\tilde q}^{p}\right]\delta_{i1i2} -$$
$$\frac{4}{81}\, g_1^4\, LF_{2,1,0}\left[m_{\tilde u}^{i1}, m_1\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{3,1,-1}\left[m_{\tilde u}^{i1}, m_1\right]\delta_{i1i2} -$$
$$\frac{4}{27}\, g_1^2\, g_3^2\, LF_{2,1,0}\left[m_{\tilde u}^{i1}, m_3\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{3,1,-1}\left[m_{\tilde u}^{i1}, m_3\right]\delta_{i1i2} -$$
$$\frac{4}{81}\, g_1^4\, LF_{2,1,0}\left[m_{\tilde u}^{i2}, m_1\right]\delta_{i1i2} + \frac{2}{81}\, g_1^4\, LF_{3,1,-1}\left[m_{\tilde u}^{i2}, m_1\right]\delta_{i1i2} -$$
$$\frac{4}{27}\, g_1^2\, g_3^2\, LF_{2,1,0}\left[m_{\tilde u}^{i2}, m_3\right]\delta_{i1i2} + \frac{2}{27}\, g_1^2\, g_3^2\, LF_{3,1,-1}\left[m_{\tilde u}^{i2}, m_3\right]\delta_{i1i2} -$$
$$\frac{5}{18}\, g_1^4\, LF_{4,1,-2}[\tilde\mu, m_1]\,\delta_{i1i2} - \frac{1}{3}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^4\, LF_{4,1,-1}[\tilde\mu, m_1]\,\delta_{i1i2} + \frac{2}{9}\, g_1^4\, LF_{5,1,-3}[\tilde\mu, m_1]\,\delta_{i1i2} +$$
$$\frac{4}{9}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^4\, LF_{5,1,-2}[\tilde\mu, m_1]\,\delta_{i1i2} - \frac{5}{6}\, g_1^2\, g_2^2\, LF_{4,1,-2}[\tilde\mu, m_2]\,\delta_{i1i2} -$$
$$m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, g_2^2\, LF_{4,1,-1}[\tilde\mu, m_2]\,\delta_{i1i2} + \frac{2}{3}\, g_1^2\, g_2^2\, LF_{5,1,-3}[\tilde\mu, m_2]\,\delta_{i1i2} +$$
$$\frac{4}{3}\, m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, g_2^2\, LF_{5,1,-2}[\tilde\mu, m_2]\,\delta_{i1i2} + \frac{1}{2}\, s_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, LF_{2,1,0}\left[\tilde\mu, m_{\tilde d}^{r}\right] -$$
$$s_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, LF_{3,1,-1}\left[\tilde\mu, m_{\tilde d}^{r}\right] + \frac{1}{2}\, s_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, LF_{4,1,-2}\left[\tilde\mu, m_{\tilde d}^{r}\right] +$$
$$\frac{1}{36}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,0}\left[\tilde\mu, m_{\tilde q}^{p}\right] - \frac{11}{36}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,-1}\left[\tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{9}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{4,1,-2}\left[\tilde\mu, m_{\tilde q}^{p}\right] + \frac{1}{2}\, s_\gamma^2\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{pi2}\, y_u^{rs}\, LF_{2,1,0}\left[\tilde\mu, m_{\tilde u}^{s}\right] -$$
$$s_\gamma^2\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{pi2}\, y_u^{rs}\, LF_{3,1,-1}\left[\tilde\mu, m_{\tilde u}^{s}\right] + \frac{1}{2}\, s_\gamma^2\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{pi2}\, y_u^{rs}\, LF_{4,1,-2}\left[\tilde\mu, m_{\tilde u}^{s}\right] +$$
$$\frac{1}{18}\, m_1\, s_\gamma\, g_1^2\, y_u^{pi2}\left(-s_\gamma\, \overline{a_u}^{pi1} + \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,0}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{36}\, m_1\, s_\gamma\, g_1^2\, y_u^{pi2}\left(-s_\gamma\, \overline{a_u}^{pi1} + \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,2,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{18}\, m_1\, s_\gamma\, g_1^2\, y_u^{pi2}\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{3,1,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{18}\, m_1\, s_\gamma\, g_1^2\, \overline{y_u}^{pi1}\left(-s_\gamma\, a_u^{pi2} + \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,0}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{36}\, m_1\, s_\gamma\, g_1^2\, \overline{y_u}^{pi1}\left(-s_\gamma\, a_u^{pi2} + \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,2,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{18}\, m_1\, s_\gamma\, g_1^2\, \overline{y_u}^{pi1}\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{3,1,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{4}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[m_1, m_{\tilde q}^{p}, \tilde\mu\right] + \frac{1}{6}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[m_1, m_{\tilde q}^{p}, \tilde\mu\right] -$$
$$\frac{1}{6}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-2}\left[m_1, m_{\tilde q}^{p}, \tilde\mu\right] - \frac{1}{6}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-1}\left[m_1, m_{\tilde q}^{p}, \tilde\mu\right] +$$
$$\frac{1}{36}\, m_1\, s_\gamma\, g_1^2\, y_u^{pi2}\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,2,1,-1}\left[m_1, m_{\tilde u}^{i1}, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{18}\, g_1^4\left(-c_\gamma^2 + s_\gamma^2\right) LF_{2,1,1,-1}\left[m_1, m_{\tilde u}^{i1}, \tilde\mu\right]\delta_{i1i2} +$$
$$\frac{1}{9}\, g_1^4\, m_1^2\left(-c_\gamma^2 + s_\gamma^2\right) LF_{2,1,1,0}\left[m_1, m_{\tilde u}^{i1}, \tilde\mu\right]\delta_{i1i2} + \frac{1}{36}\, m_1\, s_\gamma\, g_1^2\, \overline{y_u}^{pi1}\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right)$$
$$LF_{2,2,1,-1}\left[m_1, m_{\tilde u}^{i2}, m_{\tilde q}^{p}\right] + \frac{1}{18}\, g_1^4\left(-c_\gamma^2 + s_\gamma^2\right) LF_{2,1,1,-1}\left[m_1, m_{\tilde u}^{i2}, \tilde\mu\right]\delta_{i1i2} +$$
$$\frac{1}{9}\, g_1^4\, m_1^2\left(-c_\gamma^2 + s_\gamma^2\right) LF_{2,1,1,0}\left[m_1, m_{\tilde u}^{i2}, \tilde\mu\right]\delta_{i1i2} + \frac{1}{24}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,2,1,-2}\left[m_1, \tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{24}\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, (4\, m_1\, s_\gamma + 3\, \tilde\mu\, c_\gamma)\, LF_{2,2,1,-1}\left[m_1, \tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{18}\left(3\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2} + g_1^4\left(-c_\gamma^2 + s_\gamma^2\right)\delta_{i1i2}\right) LF_{2,2,1,-2}\left[m_1, \tilde\mu, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{18}\, m_1\, g_1^2\left(3\, s_\gamma\, \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\, y_u^{pi2} + m_1\, g_1^2\left(c_\gamma^2 - s_\gamma^2\right)\delta_{i1i2}\right) LF_{2,2,1,-1}\left[m_1, \tilde\mu, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{18}\left(3\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2} + g_1^4\left(-c_\gamma^2 + s_\gamma^2\right)\delta_{i1i2}\right) LF_{2,2,1,-2}\left[m_1, \tilde\mu, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{18}\, m_1\, g_1^2\left(3\, s_\gamma\, \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\, y_u^{pi2} + m_1\, g_1^2\left(c_\gamma^2 - s_\gamma^2\right)\delta_{i1i2}\right) LF_{2,2,1,-1}\left[m_1, \tilde\mu, m_{\tilde u}^{i2}\right] -$$
$$\frac{9}{4}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[m_2, m_{\tilde q}^{p}, \tilde\mu\right] -$$
$$\frac{3}{2}\, m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[m_2, m_{\tilde q}^{p}, \tilde\mu\right] + \frac{3}{2}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-2}\left[m_2, m_{\tilde q}^{p}, \tilde\mu\right] +$$
$$\frac{3}{2}\, m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-1}\left[m_2, m_{\tilde q}^{p}, \tilde\mu\right] + \frac{9}{8}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,2,1,-2}\left[m_2, \tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{3}{8}\, \tilde\mu\, c_\gamma\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, (4\, m_2\, s_\gamma + \tilde\mu\, c_\gamma)\, LF_{2,2,1,-1}\left[m_2, \tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{2}{3}\, m_3\, s_\gamma\, g_3^2\, y_u^{pi2}\left(-s_\gamma\, \overline{a_u}^{pi1} + \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,0}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{3}\, m_3\, s_\gamma\, g_3^2\, y_u^{pi2}\left(-s_\gamma\, \overline{a_u}^{pi1} + \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{2}{3}\, m_3\, s_\gamma\, g_3^2\, y_u^{pi2}\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{3,1,1,-1}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\frac{2}{3}\, m_3\, s_\gamma\, g_3^2\, \overline{y_u}^{pi1}\left(-s_\gamma\, a_u^{pi2} + \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,0}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{3}\, m_3\, s_\gamma\, g_3^2\, \overline{y_u}^{pi1}\left(-s_\gamma\, a_u^{pi2} + \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{2}{3}\, m_3\, s_\gamma\, g_3^2\, \overline{y_u}^{pi1}\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{3,1,1,-1}\left[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{3}\, m_3\, s_\gamma\, g_3^2\, y_u^{pi2}\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde u}^{i1}, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{3}\, m_3\, s_\gamma\, g_3^2\, \overline{y_u}^{pi1}\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde u}^{i2}, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{8}\, s_\gamma\, \tilde\mu\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right) LF_{2,2,1,-1}\left[m_{\tilde d}^{r}, \tilde\mu, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{8}\, s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr}\right) LF_{2,2,1,-1}\left[m_{\tilde d}^{r}, \tilde\mu, m_{\tilde q}^{s}\right] +$$
$$\frac{1}{8}\, s_\gamma\, \tilde\mu\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(-c_\gamma\, \overline{a_d}^{pr} + s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right) LF_{2,2,1,-1}\left[m_{\tilde q}^{p}, \tilde\mu, m_{\tilde d}^{r}\right] +$$
$$\frac{1}{8}\, s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(-c_\gamma\, a_d^{sr} + s_\gamma\, \tilde\mu\, y_d^{sr}\right) LF_{2,2,1,-1}\left[m_{\tilde q}^{s}, \tilde\mu, m_{\tilde d}^{r}\right] +$$
$$\frac{1}{8}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde q}^{p}\right] - \frac{1}{4}\, g_1^2\, \tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[\tilde\mu, m_1, m_{\tilde q}^{p}\right] -$$
$$\frac{1}{2}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde u}^{i1}\right] - \frac{1}{3}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[\tilde\mu, m_1, m_{\tilde u}^{i1}\right] +$$
$$\frac{1}{3}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-2}\left[\tilde\mu, m_1, m_{\tilde u}^{i1}\right] + \frac{1}{3}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde u}^{i1}\right] -$$
$$\frac{1}{2}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde u}^{i2}\right] - \frac{1}{3}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[\tilde\mu, m_1, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{3}\, g_1^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-2}\left[\tilde\mu, m_1, m_{\tilde u}^{i2}\right] + \frac{1}{3}\, m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{3,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde u}^{i2}\right] +$$
$$\frac{3}{8}\, g_2^2\, s_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,-1}\left[\tilde\mu, m_2, m_{\tilde q}^{p}\right] - \frac{3}{4}\, g_2^2\, \tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, LF_{2,1,1,0}\left[\tilde\mu, m_2, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{4}\, s_\gamma\, \tilde\mu\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right) LF_{2,1,1,0}\left[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{4}\, s_\gamma\, \tilde\mu\, y_d^{sr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(-c_\gamma\, \overline{a_d}^{pr} + s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right) LF_{3,1,1,-1}\left[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}\right] +$$
$$\frac{1}{4}\, s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr}\right) LF_{2,1,1,0}\left[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{s}\right] +$$
$$\frac{1}{4}\, s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(-c_\gamma\, a_d^{sr} + s_\gamma\, \tilde\mu\, y_d^{sr}\right) LF_{3,1,1,-1}\left[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{s}\right] -$$
$$\frac{1}{6}\, g_1^2\, y_u^{pi2}\, (m_1\, s_\gamma - \tilde\mu\, c_\gamma)\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{1,1,1,1,0}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}, \tilde\mu\right] -$$
$$\frac{1}{12}\, g_1^2\, y_u^{pi2}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}, \tilde\mu\right] -$$
$$\frac{1}{6}\, g_1^2\, \overline{y_u}^{pi1}\, (m_1\, s_\gamma - \tilde\mu\, c_\gamma)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{1,1,1,1,0}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, \tilde\mu\right] -$$
$$\frac{1}{12}\, g_1^2\, \overline{y_u}^{pi1}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, \tilde\mu\right] -$$
$$\frac{1}{4}\, \overline{y_u}^{si1}\, y_u^{pi2}\left(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\, \tilde\mu\, \overline{y_d}^{pr}\right)\left(c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr}\right) LF_{2,1,1,1,-1}\left[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, \tilde\mu\right] -$$
$$\frac{2}{9}\, g_1^2\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{i1}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{12}\, g_1^2\, y_u^{pi2}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,1,-1}\left[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{i1}, \tilde\mu\right] +$$
$$\frac{1}{12}\, g_1^2\, \overline{y_u}^{pi1}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{i2}, \tilde\mu\right] -$$
$$\frac{2}{3}\, g_3^2\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[m_{\tilde q}^{p}, m_3, m_{\tilde u}^{i1}, m_{\tilde u}^{i2}\right] +$$
$$\frac{1}{4}\, s_\gamma\, \overline{y_u}^{si1}\, y_u^{pi2}\, \overline{a_u}^{pr}\left(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr}\right) LF_{2,1,1,1,-1}\left[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, \tilde\mu\right] +$$
$$\frac{1}{4}\, \tilde\mu\, c_\gamma\left(-s_\gamma\, \overline{y_u}^{pi1}\, \overline{y_u}^{rs}\, y_u^{ri2}\, a_u^{ps} + \tilde\mu\, c_\gamma\, \overline{y_u}^{ps}\, \overline{y_u}^{ri1}\, y_u^{pi2}\, y_u^{rs}\right) LF_{2,1,1,1,-1}\left[m_{\tilde u}^{s}, m_{\tilde q}^{p}, m_{\tilde q}^{r}, \tilde\mu\right] -$$
$$\frac{1}{12}\, g_1^2\, y_u^{pi2}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,1,-1}\left[m_{\tilde u}^{i1}, m_1, m_{\tilde q}^{p}, \tilde\mu\right] -$$
$$\frac{1}{12}\, g_1^2\, \overline{y_u}^{pi1}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[m_{\tilde u}^{i2}, m_1, m_{\tilde q}^{p}, \tilde\mu\right] +$$
$$\frac{1}{12}\, g_1^2\, y_u^{pi2}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, \overline{a_u}^{pi1} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pi1}\right) LF_{2,1,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i1}\right] +$$
$$\left.\frac{1}{12}\, g_1^2\, \overline{y_u}^{pi1}\, (m_1\, s_\gamma + \tilde\mu\, c_\gamma)\left(s_\gamma\, a_u^{pi2} - \tilde\mu\, c_\gamma\, y_u^{pi2}\right) LF_{2,1,1,1,-1}\left[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}\right]\right)$$